Exam 1 - AGST 5014

Name:

2023-03-10

1. The ANOVA from a factorial randomized complete block experiment output is shown below. In this experiment, we evaluated 100 cultivars, 4 locations, and 3 replications.
a) Fill in the blanks.

Source	DF	SS	MS	F	P
Block			15		0.0000001
cultivar		138			0.0038
location			30		0.0
interaction			1.01		0.26
Error		754			
Total		1312			

b) Interpret the anova results above.

2. What are the steps required for running experiments (what would be your pipeline for running an experiment from beginning to end)? There is no clear cut on the number of steps taken. Here is the style I want you to answer:

- Wake up
- Brush my teeth
- Listen to Needtobreath
- Go to work

3. Interpret the following results:

a) Shapiro-Wilk's test

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals
## W = 0.94791, p-value < 2.2e-16
```

b) Levene's test

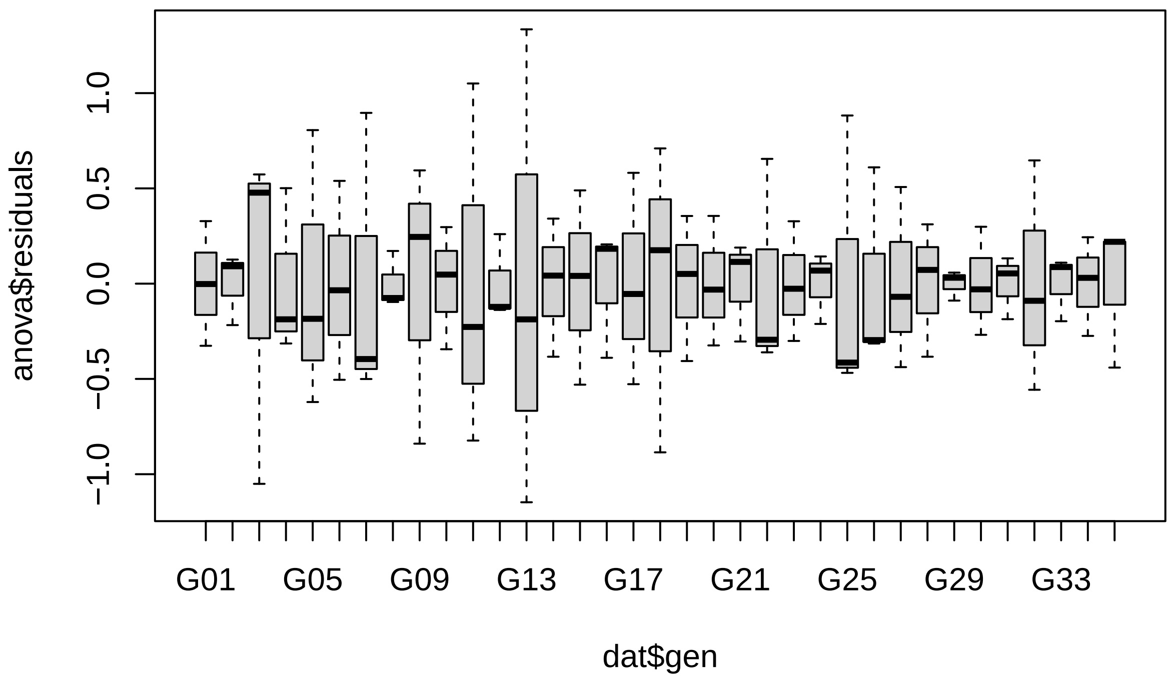
```
## Levene's Test for Homogeneity of Variance (center = median)
##       Df F value Pr(>F)
## group  9  0.8863 0.5536
##       20
```

c) Tukey 1 degree of freedom (additivity test)

```
##
## Tukey's one df test for additivity
## F = 5.1626778   Denom df = 1    p-value = 0.2639426
```

4. Can we proceed with the analysis below? Do you see anything wrong? If yes, what would you propose?

```
anova <- aov(yield ~ rep + gen, dat)
boxplot(anova$residuals ~ dat$gen)
```



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5. Interpret the following results:

```
q5_aov <- aov(Y ~ treat, data = ex1q5)
summary(q5_aov)
```

```
##             Df Sum Sq Mean Sq F value   Pr(>F)
## treat        5 1133.4  226.68   30.85 3.16e-08 ***
## Residuals   18  132.2    7.35
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
tukey(ex1q5$Y, ex1q5$treat, 18 , 132.2, alpha = 0.05)
```

```
##
## Tukey's test
## ------------------------------------------------------------------------
## Groups Treatments Means
## a      0.37    82.75
## ab     0.51    77
##  b     0.71    75
##  b     1.02    71.75
##   c    1.4     65
##   c    1.99    62.75
```

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6. **Below there is the analysis of an experiment that measured the effect of different herbicides on different crops. This was a CRD where all the treatments had the same number of replications. Based on the ANOVA table for the sums of squares type III, fill in the table for the sums of squares type I**

ANOVA Table (Type III tests)

Source	DF	SS	MS	F	P
crop	1	455	455	59.24	0.000
herbicide	3	1333	444.33	22.62	00001
interaction	3	29	9.66	3.39	0.44
Error	66	507	7.68		

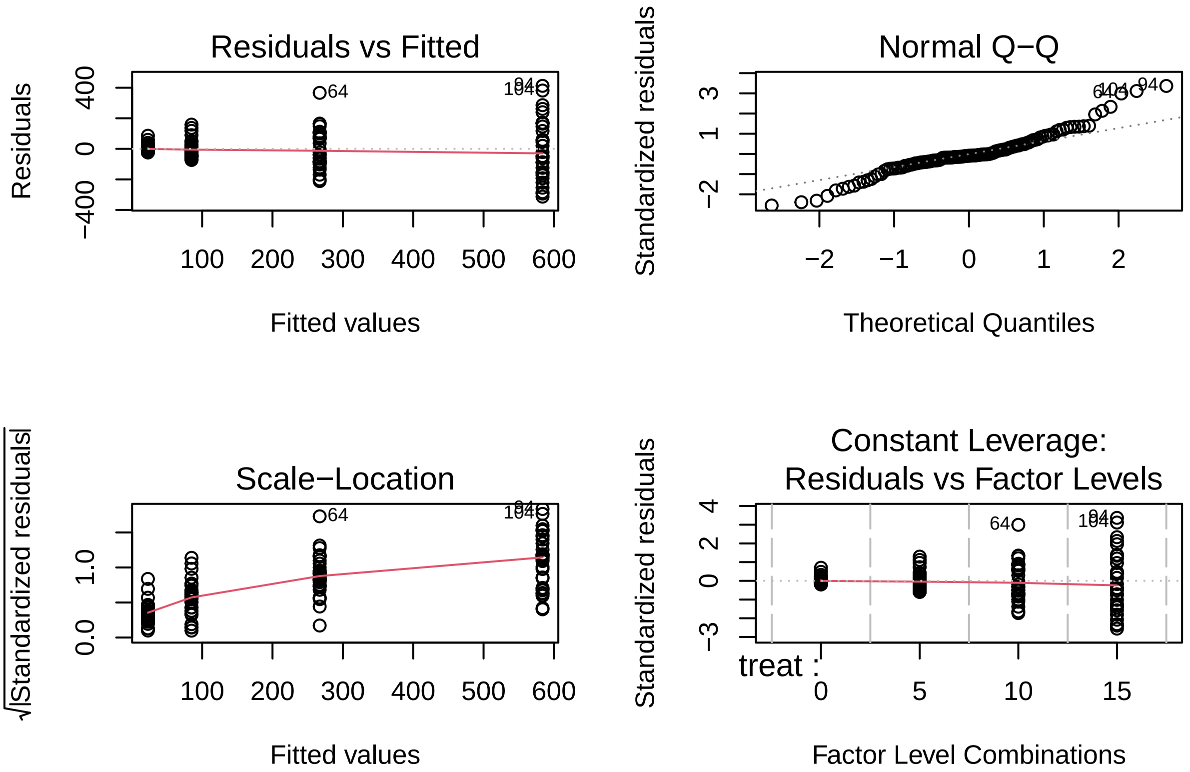
ANOVA Table (Type I tests)

Source	DF	SS	MS	F	P
crop			-		
herbicide			-		
interaction			-		
Error					

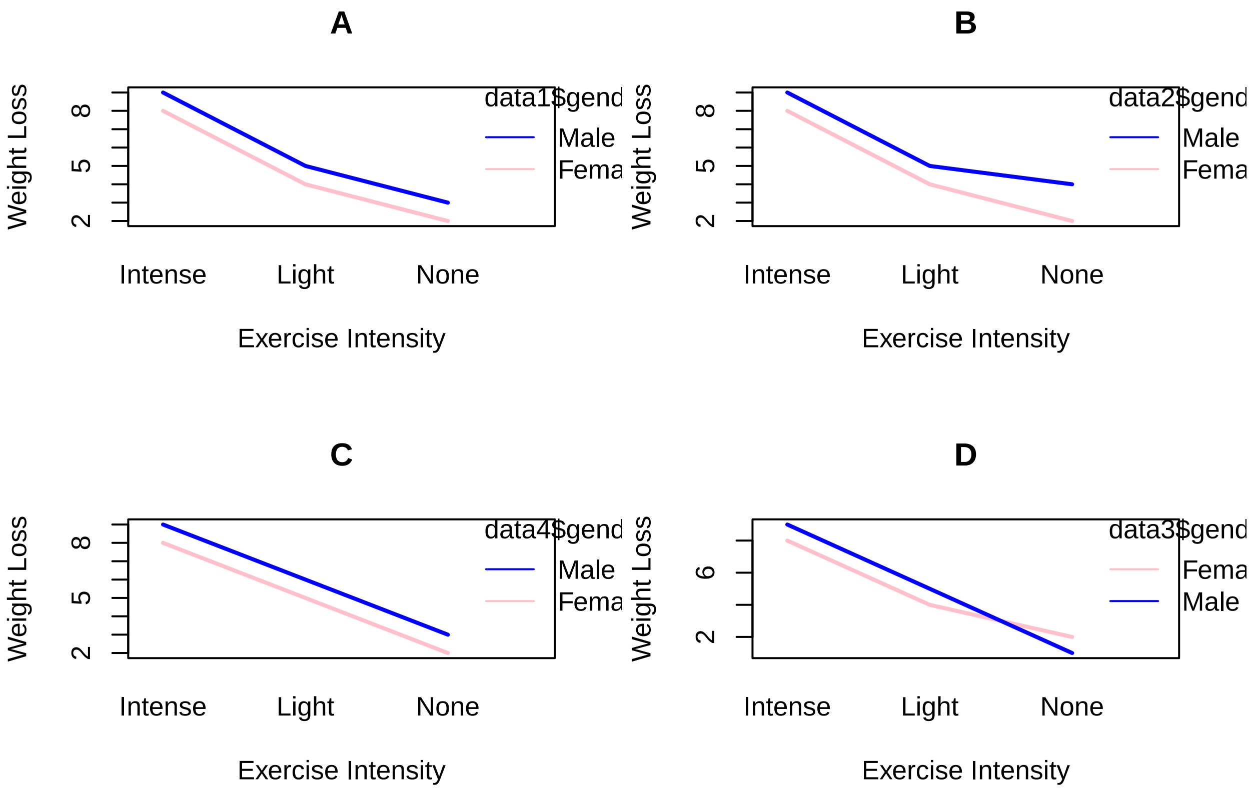
7. **Answer the following questions:**

a) What are the ANOVA requirements/assumptions?

b) Based on the diagnostic plots below, are safe using ANOVA? If not, what is the problem and what could be a solution?



8. Can we say that there is a significant interaction solely based on the interaction plot? Which of the plots below indicates a possible interaction?



9. Write the statistical model for (explain what each term is):

a) Factorial CRD

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b) RCBD

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c) CRD

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d) Factorial RCBD

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10. Answer the following questions:

a) If the treatment effect is zero, what would be the F value for treatment?

b) Testing cultivar effect (3 cultivars) in an anova. What are the null and alternative hypothesis?

c) In an anova where block is significant, but I did not account for block. The F value for treatment would be larger or smaller than expected? why?